Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Sonnenfeld, Gerald Ph.D.		
Project Title:	Suspension, the HPA Axis and Resistance to Infection		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Binghamton	**State:**	NY
Zip Code:	13902-6000	**Congressional District:**	5
Comments:	PI moved to Binghamton University, from Morehouse School of Medicine, in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-IIH00203		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project was designed to determine the effects of space flight conditions expressed in the hindlimb unloading model for rodents on resistance to infection. Additionally, the role of the neuroendocrine system in regulating resistance to infection in this system was to be determined. The specific aims of the study were: A) to expand the range of infections altered by AOH suspension. We had already shown that resistance to some infections that are not likely to be risks during space flight was altered by AOH suspension and we now determined if infections that could be a risk during space flight are affected by the suspension model; B) to determine the mechanism of alteration of resistance to infection induced by AOH suspension. Although previous studies had shown that immune responses are altered by space flight, we now extended these studies to determine the role of neuroendocrine system in regulating infections. This was carried out using two approaches. The data obtained from experiments using both approaches was integrated to allow for development of a model for the mechanism(s) of the effects of hindlimb suspension on resistance to infections. We have		

	been able to show that hindlimb-unloaded mice had decreased resistance to infection with gram negative bacteria then did control mice. We were also able to show that pretreatment with Active Hexose Correlated Compound (AHCC), a nutritional supplement, protected mice from infection. The AHCC appeared to act by enhancing immune responses in the hindlimb-unloaded mice. The same bacteria were shown to have enhanced growth in the presence of catecholamines, which suggested that the stress response occurring in the hindlimb unloaded mice could contribute to the decreased resistance to infection observed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our development of AHCC as a potential countermeasure for immune dysfunction induced by space flight has led to a new grant from the Amino Up Chemical Company to study the use of AHCC to prevent infection during severe trauma.
Task Progress:	In the past year, we have concentrated on development of Active Hexose Correlated Compound (AHCC) as a potential countermeasure for space flight-induced immune dysfunction. AHCC is a nutritional supplement that is extracted from basidiomycete mushrooms that have been shown to enhance immune responses. It is available “over-the-counter” and has been shown to be innocuous. The studies in the past year have concentrated on the mechanism of the effects of AHCC on the immune system. We had shown that treatment with (AHCC) restored resistance to infection in mice maintained under hindlimb-unloading conditions. The present study was designed to clarify the mechanisms by which AHCC enhances resistance to infection in this model. We hypothesized that oral administration of AHCC will enhance the function of the immune system, which could lead to the increased resistance to infection observed in this model. AHCC or the excipient (phosphate buffered saline) was orally administered to mice and the function of the immune system was assessed in spleen and peritoneal cells isolated from those groups. The results of the current study showed that, administration of AHCC for one week prior to and throughout the 2nd day of hindlimb-unloading period enhanced the function of the immune system assessed by spleen cell proliferation and cytokine production in spleens and nitric oxide and cytokine production in peritoneal cells. In fact, AHCC restored the function of the immune system, which was severely suppressed by hindlimb-unloading conditions, to values very close to values obtained from mice under normal control conditions. Therefore, while enhancement of the immune response in normal mice may have minimal beneficial effects on resistance to infection, restoration of the immune response to normal levels in immunosuppressed hindlimb-unloaded mice appears to be crucial for their survival. The results of the studies of the past year have contributed to answering the following critical questions: 7.03, 7.10, 7.22, 7.23, 7.26. We have shown that space flight conditions, including fluid redistribution, affect the immune system and resistance to infection and that administration of a nutritional supplement (AHCC) can prevent the development of immune dysfunction and protect against development of infection.
Bibliography Type:	Description: (Last Updated: 04/28/2011)
Articles in Peer-reviewed Journals	Aviles H, Belay T, Fountain K, Vance M, Sun B, Sonnenfeld G. "Active hexose correlated compound enhances resistance to Klebsiella pneumoniae infection in mice in the hindlimb-unloading model of spaceflight conditions." J Appl Physiol. 2003 Aug;95(2):491-6. PMID: 12692142 , Aug-2003
Articles in Peer-reviewed Journals	Aviles H, Belay T, Vance M, Sun B, Sonnenfeld G. "Active hexose correlated compound enhances the immune function of mice in the hindlimb-unloading model of spaceflight conditions." J Appl Physiol. 2004 Oct;97(4):1437-44. PMID: 15194672 , Oct-2004
Articles in Peer-reviewed Journals	Sonnenfeld G. "Animal models for the study of the effects of spaceflight on the immune system." Adv Space Res. 2003;32(8):1473-6. http://dx.doi.org/10.1016/S0273-1177(03)90383-8 PMID: 15000088 , Oct-2003
Articles in Peer-reviewed Journals	Sonnenfeld G, Butel JS, Shearer WT. "Effects of the space flight environment on the immune system." Rev Environ Health. 2003 Jan-Mar;18(1):1-17. PMID: 12875508 , Mar-2003
Articles in Peer-reviewed Journals	Aviles H, Belay T, Fountain K, Vance M, Sonnenfeld G. "Increased susceptibility to Pseudomonas aeruginosa infection under hindlimb-unloading conditions." J Appl Physiol. 2003 Jul;95(1):73-80. PMID: 12626488 , Jul-2003
Articles in Peer-reviewed Journals	Belay T, Aviles H, Vance M, Fountain K, Sonnenfeld G. "Catecholamines and in vitro growth of pathogenic bacteria: Enhancement of growth varies greatly among bacterial species." Life Sci. 2003 Aug 8;73(12):1527-35. PMID: 12865092 , Aug-2003
Books/Book Chapters	Aviles H, Sonnenfeld G. "Effects of Active Hexose Correlated Compound (AHCC) on resistance to infection of mice in the hindlimb-unloading model of space flight conditions." in "AHCC: Research and Commentary." Ed. D. Kenner. Holodigm Publishers , 2004., Jan-2004
Books/Book Chapters	Sonnenfeld G. [editor] "Experimentation with the animal model in space. Volume in Advances in Space Biology." Ed. G. Sonnenfeld. Elsevier, 2005., Jan-2005
Presentation	Sonnenfeld, G. "Space Flight, the Immune System and Resistance to Infection " Jul-2003
Presentation	Sonnenfeld, G. "Space Flight, the Immune System and Resistance to Infection " Sep-2003
Presentation	Sonnenfeld, G. "The Immune System in Space and Other Horrid Environments " Dec-2003
Presentation	Sonnenfeld, G. "The Use of Animal Models to Study the Effects of Space Flight on the Immune System " Apr-2004